

vehicles across the world. It is well known that for electric vehicles to come into more widespread use as per need, the future batteries must become even more powerful, durable, sustainable- and cost-effective.

The solution, in this regard, could be the emergence of Na-ion batteries, whose development has recently made astonishing progress. **It is foreseen that future Na-ion batteries could replace the Li-ion batteries not only in electric vehicles but also in smartphones and laptops.** Although the two alkali metals lithium and sodium are chemically very similar, sodium does not have the energy density of the comparatively rare lithium, but it is widely and cheaply available. Research activities on Na-ion battery materials have increased significantly in the past decade, with the key motive to find a reliable, safe- and cost-effective substitute for Li-ion batteries.

However, there are some key differences in the electrochemical behaviour of both battery systems. For example, the cell voltage of sodium half-cell is lower than that of lithium by 0.3V, while Na-ion conductivity is higher than Li-ion due to its polarisability. Overall, it is encouraging and helpful that most of the research carried out on Li-ion based systems has been successfully transferred to Na-ion batteries to explain and improve the fundamentals of various physio-chemical and electro-chemical phenomena within the battery system.

The verdict: Na-ion batteries have the potential to compete with Li-ion batteries

Based on different performance parameters, Na-ion batteries have the potential to compete with Li-ion batteries in three main areas of applications, starting from small to medium and large scale applications. Applications involving portable electronic devices which are presently powered by Li-ion batteries require a capacity in the range of 200-300mAhg⁻¹, with cyclic life of over 200 cycles and capacitive retention of more than 80%.

From available research data and expected improvements in electrode/

electrolyte materials, with the added advantage of lower raw material cost of Na-ion batteries, it would be safe to say that these will be strong competitors to Li-ion batteries in the near future. For a long time, research has been concentrated solely on the more powerful Li-ion batteries. Now, however, there are not only ground-breaking scientific publications, but also very promising prototypes coming of sodium-ion batteries. Some examples:

- A South Korean developed Na-ion battery has managed to handle about 500 complete charging cycles before its capacity dropped to 80%.
- A battery with a slightly different chemical structure designed by a US-Chinese research group has achieved 450 charge cycles with a similar charging capacity.
- A Chinese-prepared sodium-ion battery had a slightly lower capacity but still retained 70% of its capacity after 1,200 cycles of quick 12-minute charging.

Most sodium-ion batteries still use graphite as the anode material, making them less proficient. However, Korean researchers have now crafted new designs using modified carbon anodes which have yielded thermodynamic benefits and boosted capacity. This move promises to offset the technology's shortcomings and scientists sought to do the following:

- Promote rapid electron transport from electrolyte to active material via a porous separator.
- Make sodium ions available in large numbers to the active material across larger sites.
- Enable structured co-intercalation (ion insertion) from surface to interior.
- Maintain short diffusion paths and micro-structures.
- Increase the number of active sites.

A sodium-oxide cathode alternative to cobalt

With the current level of performance, Na-ion batteries lag behind Li-ion batteries by about twenty years, but in practice, these batteries would probably survive many more charge cycles. **Moreover, the production of the cathodes does not require rare lithium salts as simple table salt**

is sufficient to make a battery. Powerful anodes can also be produced from lignite, wood, and other biomass, making cobalt or similar rare substances unnecessary.

Challenges to be overcome

The chemistry and electro-chemistry of electrode materials for sodium-ion batteries are sufficiently different from that of their lithium counterparts, making them suitable candidates for practical batteries. The limited electrochemical stability window of water results in lower voltages and limited energy densities when aqueous electrolytes are used in Na-ion batteries.

Moreover, the disadvantages of sodium are that it is three times heavier than lithium, making Na-ion batteries also heavier. Further, sodium batteries are less powerful because they inevitably lose around ten percent of their energy density due to a 0.3V lower cell voltage, largely owing to the fact that the graphite anodes used up to now in batteries absorb too little sodium.

Recent studies by a German-Russian working group have shown that nanoscale carbon could provide a remedy to this problem. The study suggested that double layers of graphene, which is wafer-thin carbon, could store significantly more sodium atoms in the anode than the graphite used so far for this purpose.

Further technological improvements are needed to increase the performance, especially in terms of energy density as well as the technological improvement in cell-component fabrication and assembly optimisation. The future research has to be planned towards fundamental research, materials discovery, and understanding of the thermodynamic and kinetic processes governing the chemistry of these systems. In addition, the investigation to scale up Na-ion batteries is of primary importance to obtain realistic data to benchmark the progress of the technology. **EFY**



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